YOLOv12n vs YOLO11n vs YOLOv8n on COCO 2017 (val)

Abstract

This report evaluates the YOLOv12n detector against YOLO11n and YOLOv8n using the COCO 2017 validation split (5,000 images). All models are evaluated at 640px input size on a Tesla T4 GPU using Ultralytics v8.3.63 with the same dataset configuration and preprocessing pipeline. Results show YOLOv12n achieves the best overall detection accuracy with comparable latency to YOLO11n and YOLOv8n, while improving AP across small, medium, and large objects.

1. Introduction

The YOLO family focuses on real-time object detection by performing classification and localization in a single forward pass. YOLOv12 introduces attention-centric design elements intended to improve representation quality without sacrificing speed. This report measures the practical performance of YOLOv12n (the smallest YOLOv12 variant) against YOLO11n and YOLOv8n using a standardized evaluation pipeline to quantify accuracy and speed tradeoffs.

2. Approach

2.1 Models

YOLOv12n (attention-centric), YOLO11n (Ultralytics baseline), YOLOv8n (Ultralytics baseline).

2.2 Dataset

COCO 2017 validation set (5,000 images) with 80 object classes.

2.3 Evaluation Protocol

Ultralytics 8.3.63 on Tesla T4 GPU, input size 640, batch size 16, COCO AP/AR metrics.

3. Results

3.1 Overall Accuracy

Model	AP50-95	AP50	AP75	AP_S	AP_M	AP_L
YOLOv12n	0.404	0.559	0.435	0.198	0.447	0.592
YOLO11n	0.394	0.553	0.428	0.198	0.432	0.570
YOLOv8n	0.374	0.526	0.405	0.186	0.410	0.535

3.2 Recall

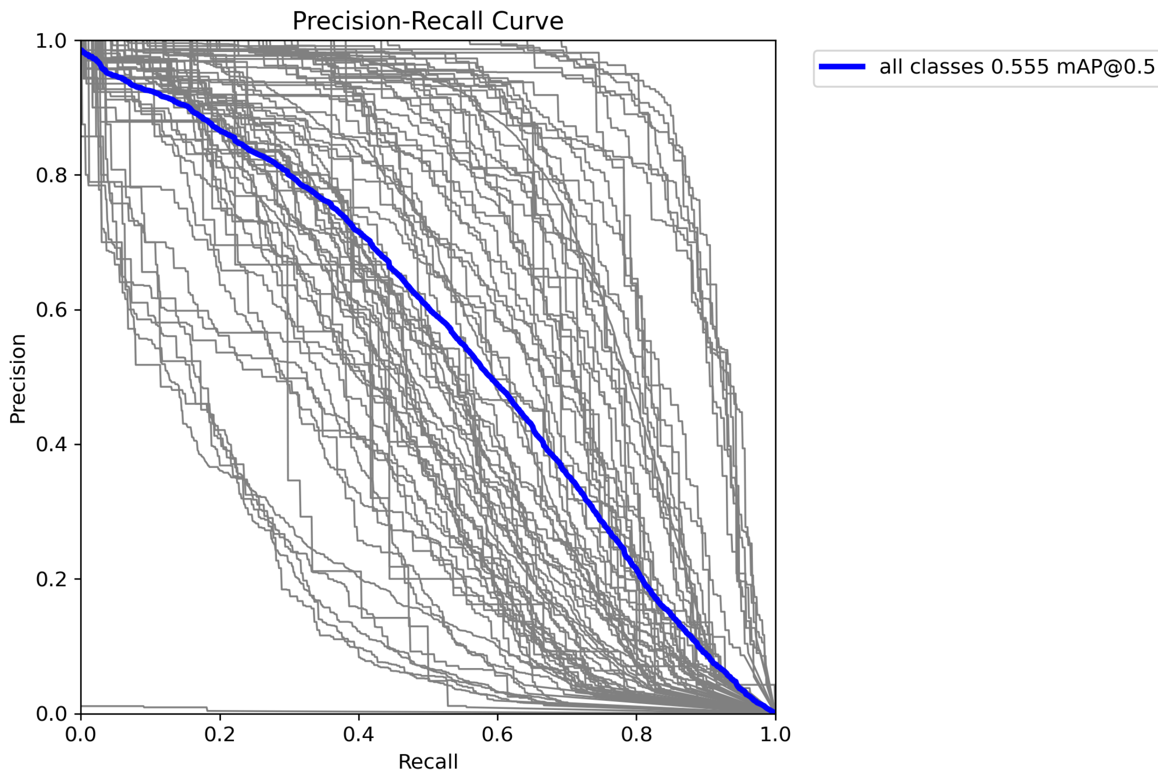
Model	AR_1	AR_10	AR_100	AR_S	AR_M	AR_L
YOLOv12n	0.329	0.549	0.605	0.368	0.672	0.795
YOLO11n	0.324	0.540	0.598	0.370	0.663	0.781

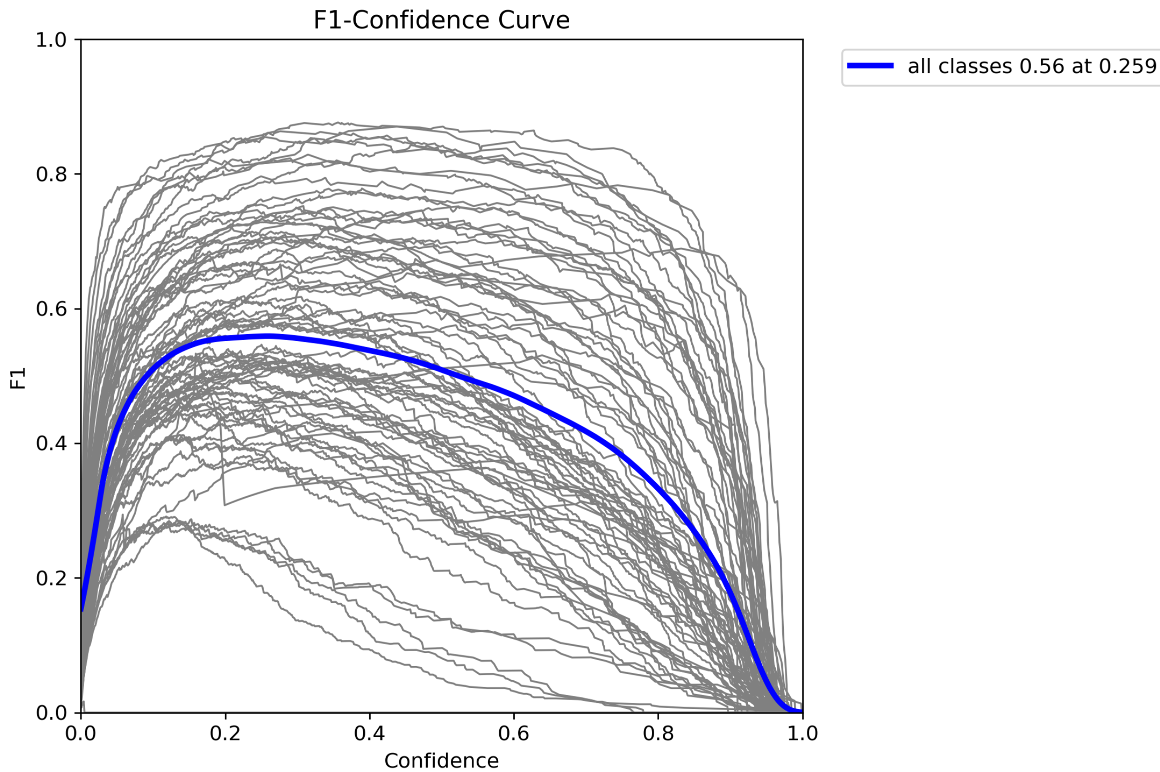
YOLOv8n	0.320	0.533	0.589	0.369	0.654	0.769

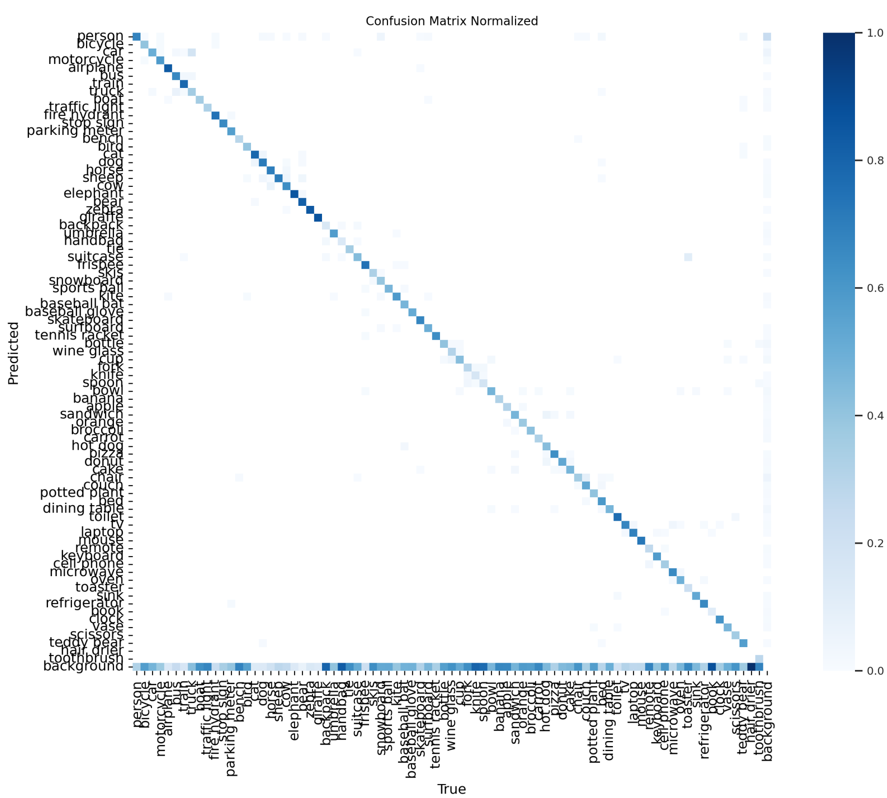
3.3 Throughput

Model	Preprocess (ms)	Inference (ms)	Postprocess (ms)
YOLOv12n	0.2	5.1	1.0
YOLO11n	0.2	2.9	0.9
YOLOv8n	0.2	3.0	1.0

3.4 Evaluation Plots (YOLOv12n)





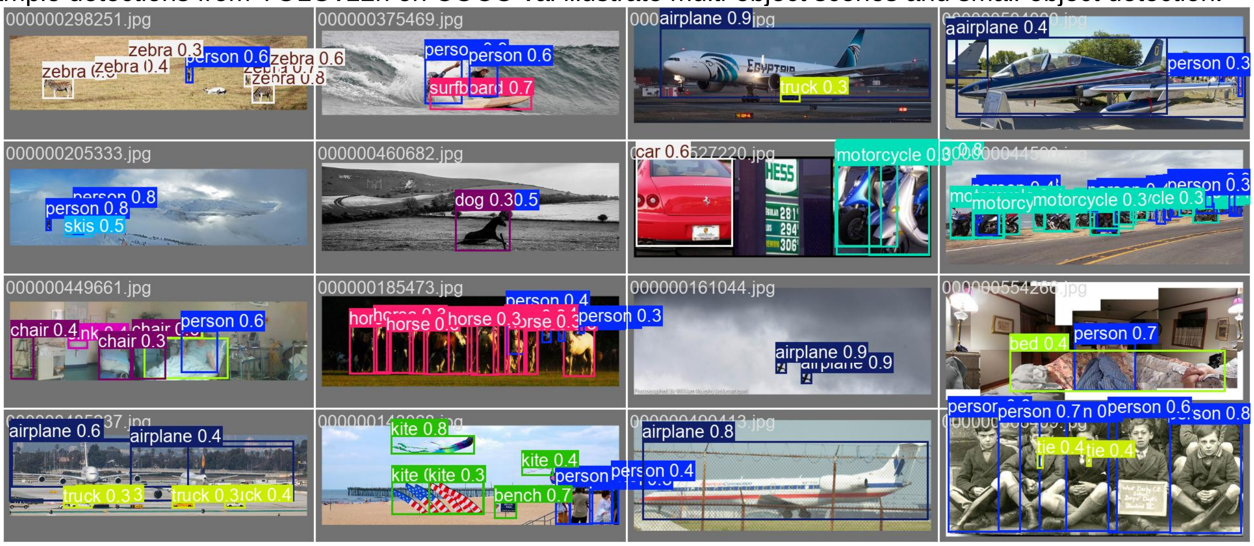


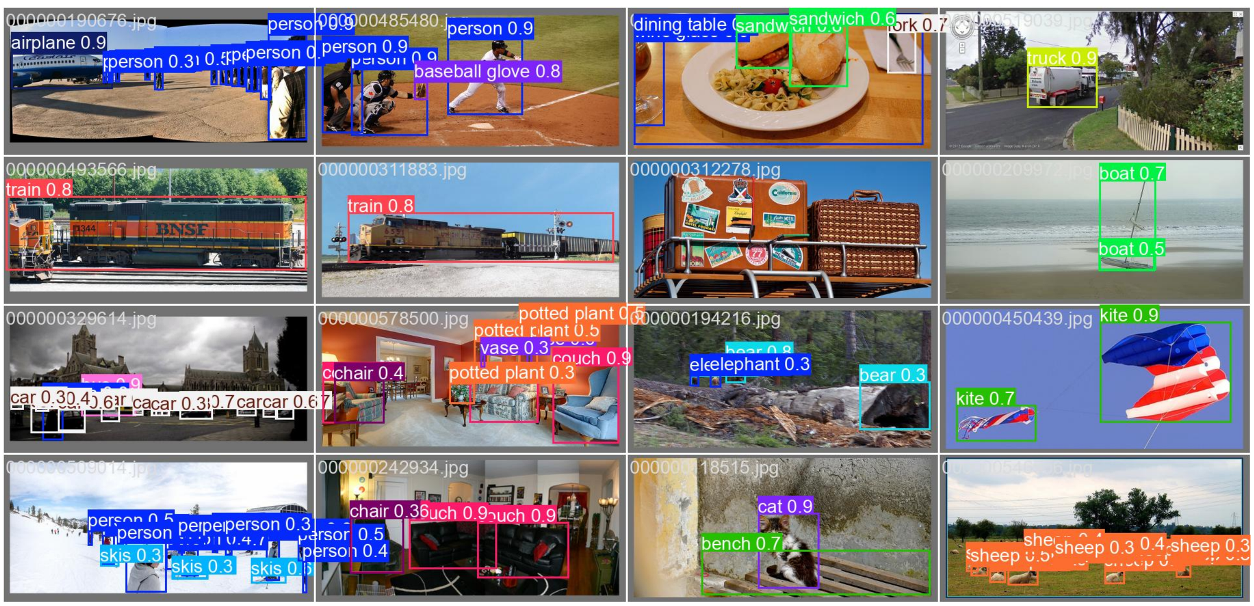
4. Discussion

YOLOv12n achieves the highest AP50-95 (0.404), a 1.0 point gain over YOLO11n and 3.0 points over YOLOv8n. Gains are consistent across object sizes, with the largest improvement on large objects. Recall improvements are modest but consistent, especially at AR_100. Inference time for YOLOv12n is higher than YOLO11n and YOLOv8n on the same hardware, indicating a small accuracy-speed tradeoff.

5. Qualitative Results

Sample detections from YOLOv12n on COCO val illustrate multi-object scenes and small-object detection.





6. Limitations

Results are limited to the COCO 2017 validation set with pretrained weights (no fine-tuning), a single input size (640), and fixed batch size. Only the smallest model variants were tested; larger models may show different tradeoffs.

7. Conclusion

YOLOv12n provides the best overall detection accuracy in this evaluation while maintaining acceptable speed on a T4 GPU. It outperforms YOLO11n and YOLOv8n across standard COCO metrics, confirming that the attention-centric design yields measurable gains without requiring additional training.

8. References

YOLOv12: Attention-Centric Real-Time Object Detectors.

Ultralytics YOLO documentation.

COCO 2017 dataset.